



Penrith Selective High School

2012

Higher School Certificate
Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Circle the correct answer to Questions 1-10 on the multiple choice answer sheet provided at the back of this paper
- Show all necessary working in Questions 11-14

Total Marks – 70

Section I Pages 2-4

10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II Pages 5-9

60 marks

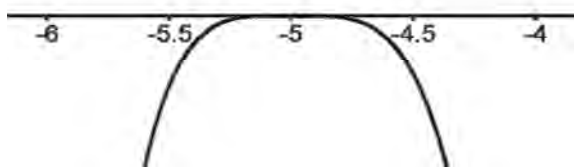
- Attempt Questions 11-14
- Allow about 1 hour 45 minutes for this section

Student Number: _____

Students are advised that this is a trial examination only and cannot in any way guarantee the content or format of the 2012 Higher School Certificate Examination.

SECTION 1: Circle the correct answer on the multiple choice answer sheet provided

- Q1. $y = f(x)$ is a linear function with slope $\frac{1}{3}$, find the slope of $y = f^{-1}(x)$.
- (A) 3 (B) $\frac{1}{3}$
(C) -3 (D) $-\frac{1}{3}$
- Q2. The interval joining the points $A(-3, 2)$ and $B(-9, y)$ is divided externally in the ratio 5 : 3 by the point $P(x, -13)$. What are the values of x and y ?
- (A) $x = -27, y = 22$ (B) $x = 27, y = 4$
(C) $x = 6, y = 12$ (D) $x = -18, y = -4$
- Q3. The value of $\lim_{x \rightarrow 0} \frac{\sin \frac{\pi x}{5} \cos \frac{\pi x}{5}}{4x} =$
- (A) $\frac{5\pi}{4}$ (B) $\frac{20}{\pi}$
(C) $\frac{4}{5\pi}$ (D) $\frac{\pi}{20}$
- Q4. Part of the graph of $y = P(x)$, where $P(x)$ is a polynomial of degree five, is shown below. This graph could be part of which of the following polynomials?



- (A) $P(x) = (x + 5)^3(x - 5)^2$ (B) $P(x) = (x + 1)(x + 5)^4$
(C) $P(x) = (x - 5)^5$ (D) $P(x) = (x - 1)^4(x + 5)$

Q5. Using $u = x^2 + 1$, the value that is equal to $\int_0^1 3x(x^2 + 1)^5 dx$ is

(A) $\frac{1}{4}$

(B) $\frac{16}{3}$

(C) $\frac{63}{4}$

(D) 32

Q6. The number N of animals in a population at time t years is given by $N = 250 + Ae^{kt}$ for constants $A > 0$ and $k > 0$. Which of the following is the correct differential equation?

(A) $\frac{dN}{dt} = k(N - 250)$

(B) $\frac{dN}{dt} = k(N + 250)$

(C) $\frac{dN}{dt} = -k(N - 250)$

(D) $\frac{dN}{dt} = -k(N + 250)$

Q7. Which of the following represents $f'(x)$ if $f(x) = \cos^{-1} 3x + x \cos^{-1} 3x$?

(A) $\cos^{-1} 3x - \frac{x+1}{\sqrt{9-x^2}}$

(B) $\cos^{-1} 3x - \frac{3(x+1)}{\sqrt{1-9x^2}}$

(C) $\cos^{-1} 3x + \frac{x-1}{\sqrt{9-x^2}}$

(D) $\cos^{-1} 3x + \frac{3(x-1)}{\sqrt{1-9x^2}}$

SECTION 2

Question 1	(15 marks) Start on a new page	Marks
a)	Solve $\frac{2x+3}{x-5} \geq 1$ for all real x .	2
b)	For $f(x) = -4 \sin^{-1}\left(\frac{2x}{3}\right)$	
i)	State the domain and range	2
ii)	Hence or otherwise, sketch $f(x)$, marking clearly any end points.	1
c)	Find the remainder when $P(x) = 5x^3 - 12x + 7$ is divided by $x + 3$.	1
d)	Find the area between the curve $y = \cos^2 2x$ and the x -axis from $x = 0$ and $x = \frac{\pi}{6}$	2
e)	Use the substitution $u = e^x$ to find: $\int \frac{dx}{e^x + 9e^{-x}}$	3
f)	i) Find the equation of the tangent to the curve $y = e^{\sin x} + x^2$ at the point where $x = \pi$.	2
	ii) Find the obtuse angle between the line $\frac{x}{2} + \frac{y}{5} = 1$ and the tangent found in part i). Give your answer to the nearest minute.	2

Question 2

(15 marks) Start on a new page

Marks

- a) A particle moves in a straight line and its position at a time t is given by $x = a \cos(9t + \theta)$. The particle is initially at the origin moving with a velocity of 15 m/s in the negative direction.
- i) Show that the particle is undergoing simple harmonic motion. **1**
 - ii) Find the period of the motion. **1**
 - iii) Find the value of the constants a and θ . **2**
 - iv) Find the position of the particle after 6 seconds, correct your answer to 2 decimal places. **1**
- b) A curve is defined by the parametric equations $x = t - 5$, $y = t^2 - 25$.
- i) Find $\frac{dy}{dx}$ in terms of t . **1**
 - ii) Find the equation of the tangent to the curve at the point where $t = -6$. **1**
 - iii) Express the equation of this curve in the simplest Cartesian form. **1**
- c) Taking $x = 3$ as the first approximation for the root of $\tan x = 1 - \log_e x$, use Newton's method to find a second approximation and correct your answer to 4 significant figures. **2**
- d) Two of the roots of the cubic polynomial $P(x) = 4x^3 - bx^2 - 64x - 16$ are reciprocals of each other, and two of the roots of $P(x)$ are of opposite signs of each other.
- i) Find the value of b . **2**
 - ii) Factorise $P(x)$ completely. **1**
- e) Find the greatest coefficient in the expansion of $(4a + 9)^{17}$. Express your answer in index form. **2**

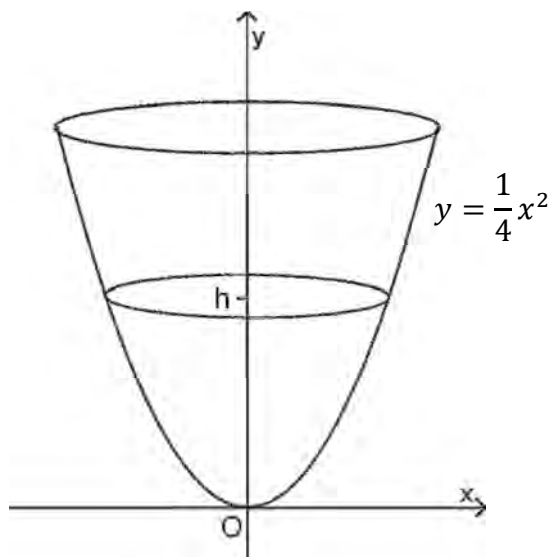
Question 3

(15 marks) Start on a new page

Marks

- a) A large industrial container is in the shape of a paraboloid, which is formed by rotating the parabola $y = \frac{1}{4}x^2$ around the y -axis.

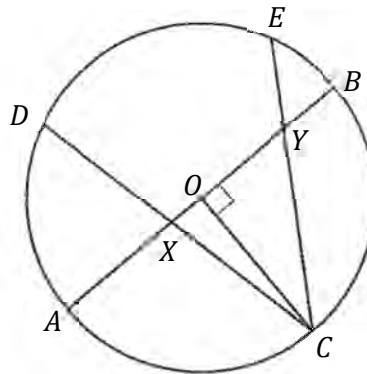
Liquid is poured into the container at the rate of 6 m^3 per minute.



- i) Prove that the volume V of liquid in the container when the depth of liquid is h , is given by $V = 2\pi h^2$. **1**
- ii) At what rate is the height of the liquid rising when the depth is 5 m? Leave your answer in exact form. **2**
- iii) If the container is 12 m high, how long will it take to fill the container? Leave your answer to two decimal places. **1**
- b) i) Use mathematical induction to prove **3**
- $$4(1^3 + 2^3 + 3^3 + \cdots + n^3) = n^2(n+1)^2$$
- for all positive integers n .
- ii) Hence find the value of $\lim_{n \rightarrow \infty} \left(\frac{1^3 + 2^3 + 3^3 + \cdots + n^3}{n^4} \right)$ **1**

Question 3 continued

- c) In the diagram below, AB is the diameter of a circle with centre O . The radius OC is drawn perpendicular to AB . The chords CD and CE intersect the diameter at the points X and Y respectively.



Copy the diagram before attempting this question.

- | | | |
|------|---|---|
| i) | Prove that $\angle CBA = \angle CAB = 45^\circ$. | 1 |
| ii) | Prove that $\angle CBD = \angle CXB$. | 2 |
| iii) | Prove that $XYED$ is a cyclic quadrilateral. | 1 |
- d) Assume that the rate at which a body cools in air is proportional to the difference between its temperature T and the constant temperature P of the surrounding air. 3

The body of the murder victim is discovered at 1am when its temperature is 33.5°C . Two hours later its temperature has fallen to 28°C . The body is cooling according to the equation $T = P + Ae^{kt}$, where t is the time in minutes and k and A are constants. If the room temperature remains constant at 22°C and assuming normal body temperature is 37°C , calculate the time the victim passed away to the nearest minute.

Question 4

(15 marks) Start on a new page

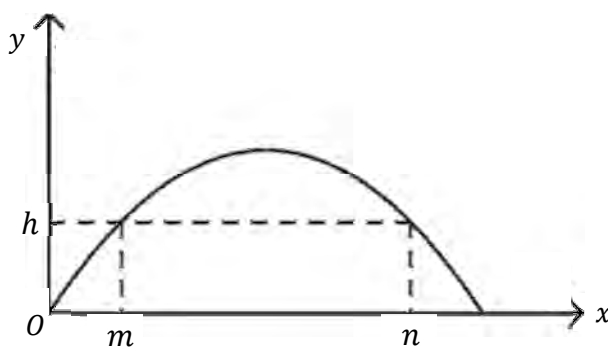
Marks

- a) i) When considering motion in a straight line, prove that 2

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \ddot{x}$$

- ii) An object moving in a straight line has an acceleration given by 2
 $\ddot{x} = x^2(4 - x^{-3})$ where x is the displacement in metres. When the object is 1 metre to the right of the origin, it has a speed of 3 m/s. Find its speed to 2 decimal places when it is 5 metres to the right of the origin.

- b) An object is projected with velocity $V \text{ ms}^{-1}$ from a point O at an angle of elevation α . Axes x and y are taken horizontally and vertically through O . The object just clears two vertical chimneys of height h meters at horizontal distance of m metres and n metres from O . The acceleration due to gravity is taken as 10 ms^{-2} and air resistance is ignored.



- i) Show that the expression of the particle after time t seconds for the 2
horizontal displacement is $x = Vt \cos \alpha$ and the vertical displacement is $y = Vt \sin \alpha - \frac{1}{2}gt^2$.

- ii) Show that $V^2 = \frac{5m^2(1 + \tan^2 \alpha)}{m \tan \alpha - h}$ 2

- iii) Show that $\tan \alpha = \frac{h(m+n)}{mn}$ 3

- c) It is given that $(1+x)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} x^k$. Show that

- i) $\sum_{k=0}^{2n} \binom{2n}{k} = 4^n$ 1

- ii) $\sum_{k=0}^{2n} \binom{2n}{k} \frac{1}{k+1} = \frac{4^{n+1} - 2}{4n+2}$ 3

-End of Paper-

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Student Number: _____

Multiple Choice Answer Sheet

- | | | | | |
|-----|---|---|---|---|
| 1. | A | B | C | D |
| 2. | A | B | C | D |
| 3. | A | B | C | D |
| 4. | A | B | C | D |
| 5. | A | B | C | D |
| 6. | A | B | C | D |
| 7. | A | B | C | D |
| 8. | A | B | C | D |
| 9. | A | B | C | D |
| 10. | A | B | C | D |

Section 1

Q1. A

$$y = mx + b$$

$$y = \frac{1}{3}x + b$$

inverse function

$$x = \frac{1}{3}y + b$$

$$y = 3x - b$$

$$\therefore m = 3 \text{ for } f^{-1}(x)$$

Q2. D

$$A(-3, 2) \quad B(-9, y)$$

ratio $-5 : 3$ (externally)

$$P(x, -13)$$

$$x = \frac{-5 \times -9 + 3 \times -3}{-5 + 3}$$

$$x = -18$$

$$-13 = \frac{-5 \times y + 3 \times 2}{-5 + 3}$$

$$26 = -5y + 6$$

$$y = -4$$

Q3. D

$$\lim_{x \rightarrow 0} \frac{\sin \frac{\pi x}{5} \cos \frac{\pi x}{5}}{4x}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{2} \sin \frac{2\pi x}{5}}{4x}$$

$$= \frac{1}{2} \times \frac{2\pi}{20}$$

$$= \frac{\pi}{20}$$

Q4. B

Q5. C

$$u = x^2 + 1 \quad du = 2x dx$$

$$x = 1 \quad u = 2$$

$$x = 0 \quad u = 1$$

$$\int_0^1 3x(x^2 + 1)^5 dx$$

$$= \frac{3}{2} \int_0^1 (x^2 + 1)^5 \times 2x dx$$

$$= \frac{3}{2} \int_1^2 u^5 du$$

$$= \frac{3}{2} \left[\frac{u^6}{6} \right]_1^2$$

$$= \frac{63}{4}$$

Q6. A

$$N = 250 + Ae^{kt}$$

$$\frac{dN}{dt} = kAe^{kt}$$

$$\frac{dN}{dt} = k(N - 250)$$

Q7. B

$$f(x) = \cos^{-1} 3x + x \cos^{-1} 3x$$

$$f'(x) = \frac{-3}{\sqrt{1-9x^2}} + \frac{-3x}{\sqrt{1-9x^2}} + \cos^{-1} 3x$$

$$f'(x) = \cos^{-1} 3x - \frac{3(x+1)}{\sqrt{1-9x^2}}$$

Q8. B

$$CL = CN = 6 \text{ cm}$$

$$LB = BM = 2 \text{ cm}$$

$$AN = AM = x \text{ cm}$$

$$(x+2)^2 + 8^2 = (x+6)^2$$

$$x^2 + 4x + 4 + 64 = x^2 + 12x + 36$$

$$8x = 32$$

$$x = 4$$

Q9. C

Maximum height when $\dot{y} = 0$

$$y = Vt \sin \varphi - \frac{1}{2}gt^2$$

$$\dot{y} = V \sin \varphi - gt$$

$$V \sin \varphi - gt = 0$$

$$t = \frac{V \sin \varphi}{g}$$

Maximum height

$$y = V \times \frac{V \sin \varphi}{g} \times \sin \varphi - \frac{1}{2}g \left(\frac{V \sin \varphi}{g} \right)^2$$

$$y = \frac{V^2 \sin^2 \varphi}{2g}$$

Q10. D

$$\left(5x^2 - \frac{3}{x} \right)^{12}$$

$$T_{r+1} = {}^{12}C_r (5x^2)^{12-r} \times \left(-\frac{3}{x} \right)^r$$

$$T_{r+1} = {}^{12}C_r \times 5^{12-r} \times (-3)^r \times x^{24-2r} \times x^{-r}$$

$$T_{r+1} = {}^{12}C_r \times 5^{12-r} \times (-3)^r \times x^{24-3r}$$

Term independent of x , so $24 - 3r = 0$

$$r = 8$$

$$T_9 = {}^{12}C_8 \times 5^4 \times (-3)^8$$

Section 2

Q1.

a)

$$\frac{2x+3}{x-5} \geq 1$$

$$(2x+3)(x-5) \geq (x-5)^2 \quad x \neq 5$$

$$2x^2 - 7x - 15 \geq x^2 - 10x + 25$$

$$x^2 + 3x - 40 \geq 0$$

$$(x+8)(x-5) \geq 0$$

$$x \leq -8, \quad x > 5$$

b) i)

$$f(x) = -4 \sin^{-1}\left(\frac{2x}{3}\right)$$

Domain:

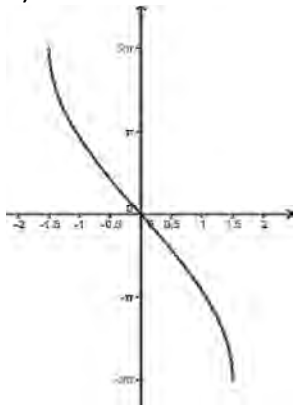
$$-1 \leq \frac{2x}{3} \leq 1$$

$$-\frac{3}{2} \leq x \leq \frac{3}{2}$$

Range:

$$-2\pi \leq y \leq 2\pi$$

ii)



c)

$P(x) = 5x^3 - 12x + 7$ is divided by $x + 3$

$$R(x) = P(-3) = 5 \times (-3)^3 - 12 \times (-3) + 7$$

$$R(x) = -92$$

d)

$$A = \int_0^{\pi/6} \cos^2 2x \, dx$$

$$A = \frac{1}{2} \int_0^{\pi/6} (\cos 4x + 1) \, dx$$

$$V = \frac{1}{2} \left[\frac{1}{4} \sin 4x + x \right]_0^{\pi/6}$$

$$V = \frac{1}{2} \left[\frac{1}{4} \times \frac{\sqrt{3}}{2} + \frac{\pi}{6} \right]$$

$$V = \frac{\sqrt{3}}{16} + \frac{\pi}{12} \text{ units}^3$$

e)

$$u = e^x$$

$$du = e^x dx$$

$$dx = \frac{1}{u} du$$

$$\int \frac{dx}{e^x + 9e^{-x}}$$

$$= \int \frac{1}{u + \frac{9}{u}} \times \frac{1}{u} du$$

$$= \int \frac{u}{u^2 + 9} \times \frac{1}{u} du$$

$$= \int \frac{du}{u^2 + 9}$$

$$= \frac{1}{3} \tan^{-1}\left(\frac{u}{3}\right) + C$$

$$= \frac{1}{3} \tan^{-1}\left(\frac{e^x}{3}\right) + C$$

f) i)

$$y = e^{\sin x} + x^2$$

$$\frac{dy}{dx} = \cos x e^{\sin x} + 2x$$

$$\text{at } x = \pi, y = 1 + \pi^2$$

$$\frac{dy}{dx} = \cos \pi e^{\sin \pi} + 2\pi$$

$$\frac{dy}{dx} = 2\pi - 1$$

Equation of tangent is

$$(y - 1 - \pi^2) = (2\pi - 1)(x - \pi)$$

$$y = (2\pi - 1)x - 2\pi^2 + \pi + \pi^2 + 1$$

$$y = (2\pi - 1)x - \pi^2 + \pi + 1$$

ii)

$$\frac{x}{2} + \frac{y}{5} = 1$$

$$5x + 2y = 10$$

$$y = -\frac{5}{2}x + 5$$

$$m_1 = -\frac{5}{2} \quad m_2 = 2\pi - 1$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan \theta = \left| \frac{-\frac{5}{2} - (2\pi - 1)}{1 + \left(-\frac{5}{2}\right) \times (2\pi - 1)} \right|$$

$$\theta = 32^\circ 31' \text{ (nearest minute)}$$

$$\text{obtuse angle} = 147^\circ 29' \text{ (nearest minute)}$$

Q2.

a) i)

$$x = a \cos(9t + \theta)$$

$$\dot{x} = -9a \sin(9t + \theta)$$

$$\ddot{x} = -81a \cos(9t + \theta)$$

$$\ddot{x} = -81x$$

As the particle's motion can be described in the form $\ddot{x} = -n^2x$, where $n = 9$, it is undergoing simple harmonic motion.

ii)

The period of the motion is $\frac{2\pi}{n} = \frac{2\pi}{9}$

iii)

$$x = a \cos(9t + \theta)$$

When $t = 0, x = 0$

$$0 = a \cos \theta$$

$$\theta = \frac{\pi}{2}$$

$$\therefore x = a \cos\left(9t + \frac{\pi}{2}\right)$$

$$\dot{x} = -9a \sin\left(9t + \frac{\pi}{2}\right)$$

When $t = 0, v = -15$

$$-15 = -9a \sin\left(\frac{\pi}{2}\right)$$

$$a = \frac{5}{3} \text{ m}$$

iv)

$$x = \frac{5}{3} \cos\left(9t + \frac{\pi}{2}\right)$$

When $t = 6$

$$x = \frac{5}{3} \cos\left(54 + \frac{\pi}{2}\right)$$

$$x = 0.93 \text{ m (2 d.p.)}$$

The particle is about 0.93 m to the right of origin.

b) i)

$$x = t - 5, \quad y = t^2 - 25$$

$$\frac{dx}{dt} = 1, \quad \frac{dy}{dt} = 2t$$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = 2t$$

ii)

$$t = -6$$

$$\frac{dy}{dx} = 2 \times -6 = -12$$

$$x = -11, \quad y = 11$$

$$y - 11 = -12(x + 11)$$

$$12x + y + 121 = 0$$

iii)

$$t = x + 5$$

$$y = (x + 5)^2 - 25$$

$$y = x^2 + 10x + 25 - 25$$

$$y = x^2 + 10x$$

c)

$$\text{Let } f(x) = \tan x + \log_e x - 1$$

$$f'(x) = \sec^2 x + \frac{1}{x}$$

$$\text{Let } x_1 = 3$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_2 = 3 - \frac{f(3)}{f'(3)}$$

$$x_2 = 3 - \frac{\tan 3 + \log_e 3 - 1}{\sec^2 3 + \frac{1}{3}}$$

$$x_2 = 3.032 \text{ (4 sig figs)}$$

d) i)

$$P(x) = 4x^3 - bx^2 - 64x - 16$$

$$P(x) = 4\left(x^3 - \frac{b}{4}x^2 - 16x - 4\right)$$

$$\text{Let the roots be } \alpha, \frac{1}{\alpha} \text{ and } \beta$$

$$\alpha \times \frac{1}{\alpha} \times \beta = 4$$

$$\beta = 4$$

$$\alpha = -4, \quad \frac{1}{\alpha} = -\frac{1}{4}$$

$$\alpha + \beta + \frac{1}{\alpha} = \frac{b}{4}$$

$$b = -1$$

ii)

$$P(x) = 4(x - 4)(x + 4)\left(x + \frac{1}{4}\right)$$

$$P(x) = (x - 4)(x + 4)(4x + 1)$$

e)

$$d) (4a + 9)^{17}$$

$$T_{k+1} = \binom{17}{k} (4a)^{17-k} (9)^k$$

$$T_{k+1} = \binom{17}{k} 4^{17-k} \cdot 9^k \cdot a^{17-k}$$

$$T_k = \binom{17}{k-1} (4a)^{17-(k-1)} (9)^{k-1}$$

$$T_k = \binom{17}{k-1} 4^{18-k} \cdot 7^{k-1} \cdot a^{18-k}$$

Compare coefficients

$$\frac{t_{k+1}}{t_k} = \frac{\binom{17}{k} 4^{17-k} \cdot 9^k}{\binom{17}{k-1} 4^{18-k} \cdot 9^{k-1}}$$

$$\frac{t_{k+1}}{t_k} = \frac{\binom{17}{k} \cdot 9}{\binom{17}{k-1} \cdot 4}$$

$$\frac{t_{k+1}}{t_k} = \frac{9}{4} \times \frac{\frac{17!}{(17-k)!k!}}{\frac{17!}{(17-(k-1))!(k-1)!}}$$

$$\frac{t_{k+1}}{t_k} = \frac{9}{4} \times \frac{(18-k)!(k-1)!}{(17-k)!k!}$$

$$\frac{t_{k+1}}{t_k} = \frac{9}{4} \times \frac{18-k}{k}$$

$$\frac{t_{k+1}}{t_k} = \frac{162-9k}{4k}$$

For the greatest coefficient,

$$\frac{t_{k+1}}{t_k} > 1$$

$$\frac{162-9k}{4k} > 1$$

$$162-9k > 4k$$

$$-13k > -162$$

$$k < 12 \frac{6}{13}$$

So the term with the greatest coefficient occurs

when $k = 12$.

$$T_{13} = \binom{17}{12} 4^{17-12} \cdot 9^{12} \cdot a^{17-12}$$

$$= \binom{17}{12} 4^5 \cdot 9^{12} \cdot a^5$$

So the greatest coefficient is $\binom{17}{12} 4^5 \cdot 9^{12}$

Q3.

a) i)

$$V = \pi \int_0^h x^2 dy$$

$$V = \pi \int_0^h 4y dy$$

$$V = \pi [2y^2]_0^h$$

$$V = 2\pi h^2$$

ii)

$$\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$$

$$\frac{dV}{dt} = 4\pi h \times \frac{dh}{dt}$$

$$\frac{dV}{dt} = 6, \quad h = 5$$

$$6 = 4\pi \times 5 \times \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{3}{10\pi} \text{ m/min}$$

iii)

$$\frac{dh}{dt} = \frac{3}{2\pi h}$$

$$t = \int_0^{12} \frac{2\pi h}{3} dh$$

$$t = \left[\frac{\pi h^2}{3} \right]_0^{12}$$

$$t = \frac{144\pi}{3} \text{ min} \approx 150.80 \text{ min}$$

b) i)

$$4(1^3 + 2^3 + 3^3 + \dots + n^3) = n^2(n+1)^2$$

Step 1: for $n = 1$

$$LHS = 4 \times 1^3 = 4$$

$$RHS = 1^2 \times (1+1)^2 = 4$$

$$LHS = RHS$$

$\therefore$ Statement is true for $n = 1$

Step 2: Assume statement is true for $n = k$

$$4(1^3 + 2^3 + 3^3 + \dots + k^3) = k^2(k+1)^2$$

Step 3: Prove statement is true for $n = k + 1$

$$\text{i.e. } 4(1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3)$$

$$= (k+1)^2(k+2)^2$$

$$LHS = 4(1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3)$$

$$LHS = 4(1^3 + 2^3 + 3^3 + \dots + k^3) + 4(k+1)^3$$

$$LHS = k^2(k+1)^2 + 4(k+1)^3 \text{ (from step 2)}$$

$$LHS = (k+1)^2(k^2 + 4(k+1))$$

$$LHS = (k+1)^2(k^2 + 4k + 4)$$

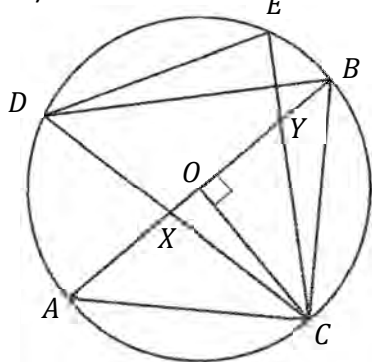
$$LHS = (k+1)^2(k+2)^2 = RHS$$

$\therefore$ Statement is true for all positive integers n by mathematical induction.

ii)

$$\begin{aligned} & \lim_{n \rightarrow \infty} \left(\frac{1^3 + 2^3 + 3^3 + \dots + n^3}{n^2} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{n^2(n+1)^2}{4n^4} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{1 + \frac{2}{n} + \frac{1}{n^2}}{4} \right) \\ &= \frac{1}{4} \end{aligned}$$

c)



i)

Join AC

$\angle CBA = \frac{1}{2} \angle AOC = 45^\circ$ (angle at the centre is twice the angle at the circumference)

$\angle CAB = \frac{1}{2} \angle AOC = 45^\circ$ (angle at the centre is twice the angle at the circumference)

$\therefore \angle CBA = \angle CAB = 45^\circ$

ii)

Join DB

$\angle DBA = \angle DCA$ (angles at the circumference standing on the same arc AD)

Let $\angle DBA = \angle DCA = \alpha$

$\angle DBC = \angle DBA + \angle ABC = \alpha + 45^\circ$ (adjacent angles)

$\angle BXC = \angle BAC + \angle DCA = \alpha + 45^\circ$ (exterior angle of $\triangle AXC$ equals to sum of the opposite interior angles)

$\therefore \angle DBC = \angle BXC$

iii)

Join DE

$\angle DBC = \angle DEC = \alpha + 45^\circ$ (angles at the circumference standing on the same arc CD)

$\therefore \angle DEC = \angle BXC = \alpha + 45^\circ$

$\therefore XYED$ is a cyclic quadrilateral as the exterior angle equals to opposite interior angle)

d)

$$T = P + Ae^{kt}$$

Let 1am be time zero

$$P = 22^\circ\text{C}, t = 0, T = 33.5^\circ\text{C}$$

$$33.5 = 22 + Ae^0$$

$$A = 11.5$$

2 hours after 1am, $t = 120, T = 28^\circ\text{C}$

$$28 = 22 + 11.5e^{120k}$$

$$\frac{6}{11.5} = e^{120k}$$

$$k = \frac{\ln\left(\frac{6}{11.5}\right)}{120}$$

Body temperature was originally 37°C

$$37 = 22 + 11.5e^{kt}$$

$$15 = 11.5e^{kt}$$

$$kt = \ln\left(\frac{15}{11.5}\right)$$

$$t = \frac{\ln\left(\frac{15}{11.5}\right)}{k}$$

$$t = -49.00859092 \dots$$

$$t \approx -49$$

$\therefore$ The victim passed away 49 min before 1am, so the time would be 12:11am in the morning.

Q4.

a) i)

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{1}{2} \times 2 \times v \times \frac{dv}{dx}$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = v \frac{dv}{dx}$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{dx}{dt} \times \frac{dv}{dx}$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{dv}{dt}$$

$$\therefore \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{d^2 x}{dt^2}$$

ii)

$$\ddot{x} = x^2(4 - x^{-3})$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = x^2(4 - x^{-3})$$

$$\frac{1}{2} v^2 = \int \left(4x^2 - \frac{1}{x} \right) dx$$

$$\frac{1}{2} v^2 = \frac{4x^3}{3} - \ln x + C$$

$$v^2 = \frac{8x^3}{3} - 2 \ln x + C$$

When $x = 1$, $v = 3$

$$9 = \frac{8}{3} + C$$

$$C = 6\frac{1}{3}$$

$$\therefore v^2 = \frac{8x^3}{3} - 2 \ln x + 6\frac{1}{3}$$

$$x = 5$$

$$v^2 = \frac{8}{3} \times 5^3 - 2 \ln 5 + 6\frac{1}{3}$$

$$v = \pm 18.34 \text{ m/s (2 d.p.)}$$

The speed is 18.34 m/s when the particle is 5 metres to the right of the origin.

b) i)

Horizontal

$$\ddot{x} = 0$$

$$\dot{x} = V \cos \alpha$$

$$x = Vt \cos \alpha$$

Vertically

$$\ddot{y} = -g$$

$$\dot{y} = V \sin \alpha - gt$$

$$y = Vt \sin \alpha - \frac{1}{2}gt^2$$

ii)

at $x = m$, $y = h$, $g = 10$

$$m = Vt \cos \alpha$$

$$t = \frac{m}{V \cos \alpha}$$

$$h = V \times \frac{m}{V \cos \alpha} \times \sin \alpha - \frac{1}{2} \times 10 \times \left(\frac{m}{V \cos \alpha} \right)^2$$

$$h = m \tan \alpha - \frac{5m^2}{V^2 \cos^2 \alpha}$$

$$h = m \tan \alpha - \frac{5m^2(\tan^2 \alpha + 1)}{V^2}$$

$$5m^2(\tan^2 \alpha + 1) = V^2(m \tan \alpha - h)$$

$$V^2 = \frac{5m^2(\tan^2 \alpha + 1)}{m \tan \alpha - h}$$

iii)

$$V^2 = \frac{5n^2(\tan^2 \alpha + 1)}{n \tan \alpha - h}$$

$$\frac{5m^2(\tan^2 \alpha + 1)}{m \tan \alpha - h} = \frac{5n^2(\tan^2 \alpha + 1)}{n \tan \alpha - h}$$

$$\frac{m \tan \alpha - h}{m^2} = \frac{n \tan \alpha - h}{n^2}$$

$$m^2(n \tan \alpha - h) = n^2(m \tan \alpha - h)$$

$$m^2 n \tan \alpha - m^2 h = n^2 m \tan \alpha - n^2 h$$

$$m^2 n \tan \alpha - n^2 m \tan \alpha = m^2 h - n^2 h$$

$$(m^2 n - n^2 m) \tan \alpha = m^2 h - n^2 h$$

$$\tan \alpha = \frac{m^2 h - n^2 h}{m^2 n - n^2 m}$$

$$\tan \alpha = \frac{(m^2 - n^2)h}{mn(n - m)}$$

$$\tan \alpha = \frac{(m + n)(m - n)h}{mn(n - m)}$$

$$\tan \alpha = \frac{h(m + n)}{mn}$$

c) i)

$$(1+x)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} x^k$$

Let $x = 1$

$$2^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} 1^k$$

$$4^n = \sum_{k=0}^{2n} \binom{2n}{k}$$

ii)

Integrate both sides

$$\frac{(1+x)^{2n+1}}{2n+1} = \sum_{k=0}^{2n} \binom{2n}{k} \frac{x^{k+1}}{k+1} + C$$

Let $x = 0$

$$\frac{1}{2n+1} = C$$

$$\frac{(1+x)^{2n+1}}{2n+1} = \sum_{k=0}^{2n} \binom{2n}{k} \frac{x^{k+1}}{k+1} + \frac{1}{2n+1}$$

$$\sum_{k=0}^{2n} \binom{2n}{k} \frac{x^{k+1}}{k+1} = \frac{(1+x)^{2n+1}}{2n+1} - \frac{1}{2n+1}$$

Let $x = 1$

$$\sum_{k=0}^{2n} \binom{2n}{k} \frac{1}{k+1} = \frac{2^{2n+1}}{2n+1} - \frac{1}{2n+1}$$

$$\sum_{k=0}^{2n} \binom{2n}{k} \frac{1}{k+1} = \frac{2^{2n+1} - 1}{2n+1}$$

$$\sum_{k=0}^{2n} \binom{2n}{k} \frac{1}{k+1} = \frac{2 \times 2^{2n} - 1}{2n+1} \times \frac{2}{2}$$

$$\sum_{k=0}^{2n} \binom{2n}{k} \frac{1}{k+1} = \frac{4 \times 4^n - 2}{4n+2}$$

$$\sum_{k=0}^{2n} \binom{2n}{k} \frac{1}{k+1} = \frac{4^{n+1} - 2}{4n+2}$$